Status	Finished
Started	Sunday, 25 October 2026, 11:54 AM
Completed	Sunday, 25 October 2026, 11:59 AM
Duration	5 mins 13 secs
Grade	**13.00** out of 15.00 (**86.67**%)

Question **1**

Complete

Mark 1.00 out of 1.00

What is mean by stable sorting algorithm?

a. A sorting algorithm is stable if it preserves the order of non-duplicate keys

b. A sorting algorithm is stable if it preserves the order of all keys

c. A sorting algorithm is stable if it doesn't preserver the order of duplicate keys

d. A sorting algorithm is stable if it preserves the order of duplicate keys

The correct answer is:
A sorting algorithm is stable if it preserves the order of duplicate keys

Question **2**

Complete

Mark 1.00 out of 1.00

What is the primary advantage of the divide-and-conquer approach in sorting algorithms?

a. It allows for efficient parallel processing and sorting of data

b. It simplifies the sorting process by using only one pass

c. It only works on small datasets

d. It avoids the need for recursion

The correct answer is: It allows for efficient parallel processing and sorting of data

Question **3**

Complete

Mark 1.00 out of 1.00

In Merge Sort, what happens after the two halves of the list are sorted?

a. They are compared element by element
b. They are combined to form a single sorted list
c. They are split again into smaller sublists
d. They are discarded

The correct answer is: They are combined to form a single sorted list

Question **4**

Complete

Mark 1.00 out of 1.00

Which of the following is not an in-place sorting algorithm?

a. Selection sort
b. Heap sort
c. Merge sort
d. Quick sort

The correct answer is:
Merge sort

Question **5**

Complete

Mark 1.00 out of 1.00

Which algorithm typically follows a divide-and-conquer structure?

a. Merge Sort
b. Linear Search
c. Binary Search
d. Bubble Sort

The correct answer is: Merge Sort

Question **6**

Complete

Mark 1.00 out of 1.00

What is one advantage of sorting a list before performing a search operation?

- a. It allows for faster searching
- b. It has no effect on the search operation
- c. It increases the number of comparisons needed
- d. It makes the search operation slower

The correct answer is: It allows for faster searching

Question **7**

Complete

Mark 1.00 out of 1.00

In Merge Sort, what happens after dividing the input into smaller parts?

- a. Each part is searched for a specific element
- b. The parts are merged without sorting
- c. The parts are ignored
- d. Each part is sorted independently

The correct answer is: Each part is sorted independently

Question **8**

Complete

Mark 0.00 out of 1.00

Very slow way of sorting is_____

- a. Bubble sort
- b. Heap sort
- c. Insertion sort
- d. Quick sort

The correct answer is:
Insertion sort

Question **9**

Complete

Mark 1.00 out of 1.00

What is one of the first steps in a divide-and-conquer algorithm like Merge Sort?

a. Sorting the entire list sequentially
b. Dividing the input into smaller subproblems
c. Combining sorted sublists
d. Comparing each element with the others

The correct answer is: Dividing the input into smaller subproblems

Question **10**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm involves comparing elements and swapping adjacent items that are out of order?

a. Merge Sort
b. Bubble Sort
c. Binary Search
d. Linear Search

The correct answer is: Bubble Sort

Question **11**

Complete

Mark 0.00 out of 1.00

Two-way merge sort algorithm is used to sort the following elements in ascending order.
200,470,150,80,90,40,400,300,120,70
What is the order of these elements after second pass of the merge sort algorithm?

a. 40,80,90,150,200,300,400,470,70,120
b. 200,470,80,150,40,90,300,400,70,120
c. 40,70,80,90,120,150,200,300,400,470
d. 80,150,200,470,40,90,300,400,70,120

The correct answer is:
80,150,200,470,40,90,300,400,70,120

Question **12**

Complete

Mark 1.00 out of 1.00

Which of the following best describes the process of Merge Sort?

- a. It divides the list into two halves, sorts each half, and then merges them
- b. It compares adjacent elements and swaps them if necessary
- c. It repeatedly finds the minimum element and moves it to the sorted part of the list
- d. It builds a sorted array one element at a time

The correct answer is: It divides the list into two halves, sorts each half, and then merges them

Question **13**

Complete

Mark 1.00 out of 1.00

What is a key disadvantage of Bubble Sort compared to more advanced algorithms like Merge Sort?

- a. Bubble Sort does not guarantee sorted order
- b. Bubble Sort cannot handle duplicate elements
- c. Bubble Sort is difficult to implement
- d. Bubble Sort is less efficient for large lists

The correct answer is: Bubble Sort is less efficient for large lists

Question **14**

Complete

Mark 1.00 out of 1.00

Why is sorting important for selection operations?

- a. It slows down the process
- b. It makes the data unsorted
- c. It makes it easier to select items based on their relationship to the rest of the items
- d. It complicates the selection of items

The correct answer is: It makes it easier to select items based on their relationship to the rest of the items

Question **15**

Complete

Mark 1.00 out of 1.00

What type of problems can sorting help solve efficiently?

a. Duplicates
b. All of the above
c. Searching
d. Selection

The correct answer is: All of the above